

Raunak Kumar

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Applied scientist training LLM agents with RL, with a Cornell PhD in online learning and bandits. My work spans convergence proofs and regret bounds to graders, training infrastructure, and run debugging, and my current research applies that background to how RL training should spend its compute.

EXPERIENCE

Senior Applied Scientist, Microsoft (Turing)

Redmond, WA · Sep 2024 – Present

RL post-training for Microsoft Copilot

co-technical lead, 2025 – present

- Led RL post-training for tool-using language models on multi-step email and calendar tasks, spanning environment/data design, reward modeling, training, and evaluation; trained agents against state-changing tools, requiring side-effect-aware grading and environment reset mechanisms.
- Developed an iterative training methodology driven by rollout-level failure analysis, decomposing errors across retrieval, tool use, evidence extraction, and response synthesis and using those diagnoses to redesign graders, data, and training recipes.
- Improved aggregate evaluation score by 56% relative and high-quality task pass rate by 2.2x over the base model on held-out offline validation sets; in offline product evaluations built on a different setup and tenants, the post-trained model outperformed all comparators, including substantially larger models, by 5–8 points relative, and retained its advantage under unseen tool configurations.
- Integrated the resulting recipe and checkpoints into a broader multi-domain post-training effort.

Variance-aware credit assignment for policy-gradient RL

technical lead, 2025 – present (part-time)

- Originated as a solo project at an internal hackathon: a variance-aware credit assignment method that composes with any policy-gradient algorithm, extending conditional-rollout value estimation (à la VinePPO) with compute allocated in proportion to conditional variance.
- Designed the algorithm and built the end-to-end proof of concept in one week; scoped and supervised an intern on the follow-up.
- On math reasoning, matches uniform-allocation credit assignment at 15–30% fewer GPU-hours, and keeps improving after GRPO's validation accuracy plateaus. Extension to tool-using agents in progress.
- Sep 2024 – mid 2025: ranking and retrieval for enterprise search in Microsoft 365 Copilot Chat and Search.

Graduate Research Assistant, Cornell University

Ithaca, NY · 2019 – 2024

- Online decision-making with Robert D. Kleinberg and Sarah Dean: extended online convex optimization and the bandits with knapsacks problem to settings with unbounded memory and non-monotonic resource use (two NeurIPS, one EC/JMLR).
- Network science with Austin R. Benson: neural models for temporal graphs and hypergraphs, and fast deterministic and randomized sampling algorithms for weighted subgraph retrieval in large graphs (WSDM).

Data & Applied Scientist Intern, Microsoft (MSAI)

Redmond / Bellevue · Summers 2020–23

- **2023** – Grounded LLMs in Office 365 data for Microsoft Business Chat, formulated as natural-language-to-Cypher translation over a graph database; developed iterative prompt construction and refinement.
- **2022** – Scaled Graphormer for link prediction on large graphs; simple methods gave large gains over baselines, and identified the general mechanism behind them. Also worked on stable node embeddings and fusing content with interaction signals.
- **2021** – Related-entity prediction on graph-structured data; built a reusable experimental pipeline and contributed to an internal GNN framework.
- **2020** – Topic suggestion; unsupervised and supervised methods with significant precision and recall gains, plus evaluation metrics and visualizations for error analysis.

Undergraduate Research Assistant, University of British Columbia

Vancouver, BC · 2017 – 2018

- Optimization for machine learning with Mark Schmidt: proved the first non-asymptotic convergence rate for the expectation maximization (EM) algorithm and developed improved EM variants (AISTATS best paper).

Software Engineering Intern, Google

Mountain View, CA · Summer 2016

- Built an extensible trigger utility in C++ for a distributed datastore, used internally to retire one billion triggers from storage.

SELECTED PUBLICATIONS

- [1] Learning in Budgeted Auctions with Spacing Objectives
G. Fikioris, R. Kleinberg, Y. Kolumbus, R. Kumar, Y. Mansour, É. Tardos **EC 2025** · **JMLR 2026**
- [2] Online Convex Optimization with Unbounded Memory
R. Kumar, S. Dean, R. Kleinberg **NeurIPS 2023**
- [3] Non-monotonic Resource Utilization in the Bandits with Knapsacks Problem
R. Kumar, R. Kleinberg **NeurIPS 2022**
- [4] Homeomorphic-Invariance of EM: Non-Asymptotic Convergence in KL Divergence for Exponential Families via Mirror Descent
F. Kunstner, R. Kumar, M. Schmidt **AISTATS 2021** · *Best Paper Award*
- [5] Retrieving Top Weighted Triangles in Graphs
R. Kumar*, P. Liu*, M. Charikar, A. R. Benson **WSDM 2020** · *equal contribution*
- [6] Convergence Rate of Expectation-Maximization
R. Kumar, M. Schmidt **NeurIPS 2017 OPT Workshop**

EDUCATION

Ph.D. in Computer Science, Cornell University Ithaca, NY · 2018 – 2024

- Advisors: Robert D. Kleinberg and Sarah Dean.
- Thesis: *Learning from Interactions via Online Decision-making and Network Science*.
- Focus: online convex optimization with memory, bandits with knapsacks, network science.

B.Sc. in Computer Science, with distinction, University of British Columbia Vancouver, BC · 2013 – 2018

- Advisor: Mark Schmidt.
- A+ graduating average and undergraduate research on optimization for machine learning.

AWARDS & HONORS

Research	AISTATS Best Paper Award (2021); CRA Outstanding Undergraduate Researcher Award, Honorable Mention (2018).
Fellowships	NSERC Postgraduate Scholarship, Doctoral (2020); NSERC Alexander Graham Bell Canada Graduate Scholarship, Doctoral (2020, declined); Cornell University Fellowship (2018); NSERC Canada Graduate Scholarship, Master's (2018, declined); Rick Sample Memorial Research Scholarship (2018); NSERC Undergraduate Student Research Award (2017).
Teaching	Cornell CS Teaching Assistant Award (2019); UBC CS Teaching Assistant Award (2017).
Scholarships	UBC Science Scholar / Dean's Honors List (2013–17); UBC Chancellor's Scholar Award (2013); Trek Excellence Scholarship (2014); J. Fred Muir Memorial Scholarship in Science (2014).
Contests	ACM-ICPC Pacific NW Regionals — 13 th (2016) and 15 th (2017) in Div. 1, 6 th in Div. 2 (2015); coached teams to 1 st and 3 rd in Div. 2 (2017). Microsoft College Code Competition — 2 nd (2017), 6 th (2016). NAIPC Open, USA/Canada — 15 th (2017), 27 th (2016). SFU Winter Contest — 8 th (2017). Calgary Collegiate, Div. 2 — 3 rd (2015).

TEACHING & SERVICE

Teaching	Departmental teaching awards at both Cornell and UBC. As an undergraduate, co-designed and co-taught CPSC 490 <i>Problem Solving Seminar</i> , a full 3-credit course on advanced algorithms and data structures, including lectures, notes and assignments. TA for machine learning (graduate and undergraduate), algorithms, operating systems and computer systems across both departments; led and coordinated TA teams at Cornell.
Reviewing	Reviewer or program committee for NeurIPS (2023, 2025), ICLR (2025, 2026), L4DC (2023, 2024), AAAI (2023), ITCS (2022), COLT (2020), and the NeurIPS <i>What Can('t) Transformers Do?</i> workshop (2025).
Leadership	President, Cornell CS Graduate Organization (2020–21); Special Representative for Waste Management (2019–20).
Outreach	Mentored at the Cornell High School Programming Contest (2021, 2023), Girls' Adventures in Math, and Expand Your Horizons.